

TEST PROCEDURE

Digital Twin Test Procedure

Team Name:	ROFASO
Project Name:	D.A.P. System
System Name:	Digital Twin
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Test	Result	Summary
Overall Result	PASS / FAIL	
Conducted By		
Witnessed By		
Notes		

Contents

1	Test Objective	2
2	Equipment Required	2
3	Scope	2
4	Test Setup	3
5	Test Procedure	3
5.1	Setup	3
5.2	Running the Test code	5
6	Anomalies and Deviations	7
7	Shutdown Procedure	8
7.1	Shutdown	8
8	Sign-Off	8

1. Test Objective

The following procedure aims to test the ability of the Digital Twin (DT) to take in dive files, locate the needed parameters in the files and output the appropriate simulated data. The DT must be able to simulate expected behaviors for nominal and off-nominal Seaglider geometries, including broken wing, broken rudder and no damage cases. The requirements being verified in this test procedure are the following:

- DTR-1: The Digital Twin shall output orientation (pitch angle, roll angle, and heading) with average angular error no greater than 19.8% relative to the min/max delta angle from onboard-recorded data.
- DTR-2: The Digital Twin shall output vertical position with percent error no greater than 6.6% of maximum mission depth relative to onboard-recorded depth.
- DTR-3: The Digital Twin shall output vertical velocity with percent error no greater than 3.3% of the min/max delta vertical velocity relative to onboard-recorded vertical velocity.

2. Equipment Required

Table 1: List of equipment

Equipment	Description	Qty
PC	Needed to run the software	1
MATLAB Software	R2024b verison or above, includes Aerospace Package	1
Digital Twin Test folder	Includes NC files, Log files, and all functions	1

3. Scope

The Digital Twin (DT) is a simulated model of the Seaglider's dynamic behavior that replicates its motion and response throughout a dive. Using actuator states from the variable buoyancy device (VBD), battery roll and pitch systems, mission data, and environmental parameters, the DT computes vehicle forces, moments, velocities, and motion states. The model also incorporates vehicle geometry, inertial and mass properties, and hydrodynamic coefficients that define fluid interaction. Within the D.A.P. system, the DT generates simulated estimates of the Seaglider's orientation, angular rates, and position, allowing comparison with recorded mission data to identify faults or performance deviations.

This test evaluates the ability of the Digital Twin to accurately simulate Seaglider behavior across multiple vehicle configurations. To conduct the test, mission data from the Edmonds 4/17/2026 dives will be injected into the Digital Twin, with the simulation configured to match the Seaglider configuration used during each mission. For each test, selected parameters from the Digital Twin (DT) output state vectors will be extracted and compared with the mission data recorded by the Seaglider, including depth, vertical speed, pitch angle,

roll angle, and heading. The pass criteria for each requirement are based on the allowable error magnitude associated with the specific parameter being evaluated. A test is considered a pass if each evaluated parameter remains within its corresponding allowable error margin.

4. Test Setup

The following procedure will be repeated for all test combinations described in Table 2. The table specifies which mission datasets will be run with each Seaglider configuration.

Table 2: Mission Dive and Corresponding Seaglider Configuration

Test ID	Mission Dive	Seaglider Configuration	Log & NC Files	Expected Output	Pass Criteria
NOM-1	Edmonds Dive 5	Nominal	p1940005.log p1940005.nc	Graphs of the Mission vs. Simulated data	Sim values are within their error magnitudes
FLT-1	Edmonds Dive 8	50% Missing Rudder	p1940008.log p1940008.nc	Graphs of the Mission vs. Simulated data	Sim values are within their error magnitudes
FLT-2	Edmonds Dive 11	50% Missing Wing	p19400011.log p19400011.nc	Graphs of the Mission vs. Simulated data	Sim values are within their error magnitudes

Table 3: Output comparisons against time in seconds.

Mission Data vs Digital Twin Data
Depth
Vertical Velocity
Pitch Angle
Roll Angle
Heading

5. Test Procedure

5.1. Setup

1. Open [UW Seaglider D.A.P. Github](#). OK? ____
2. From the [UW Seaglider D.A.P. Github](#), download the Digital Twin Test Folder of code function into MATLAB folder. OK? ____

 Seaglider_DAP_DT_Test_50_rudder		5/17/2026 12:10 PM	WinRAR ZIP archive	184 KB
 Seaglider_DAP_DT_Test_50_wing		5/17/2026 12:10 PM	WinRAR ZIP archive	197 KB
 Seaglider_DAP_DT_Test_nominal		5/17/2026 12:10 PM	WinRAR ZIP archive	165 KB

Figure 1: Downloaded Test Case MATLAB files

3. Open MATLAB PC application OK? ____
4. Open the D.A.P main function from the test folder, confirm that the following are in your MATLAB Files tab:
 - (a) parametersV3.m OK? ____
 - (b) NC_File_Unpacker.m OK? ____
 - (c) Log_File_Unpacker.m OK? ____
 - (d) dynamicsV4.m OK? ____
 - (e) Digital_Twin.m OK? ____
 - (f) DAP_Main.m OK? ____
 - (g) Create_Mission_Data_Matrix.m OK? ____
 - (h) coefficientsV4_nominal.m OK? ____
 - (i) coefficientsV4_50wing.m OK? ____
 - (j) coefficiteentsV4_50rudder.m OK? ____
 - (k) coefficientsV4_25wing.m OK? ____
 - (l) p1940005.log, p1940005.nc OK? ____
 - (m) p1940008.log, p1940008.nc OK? ____
 - (n) p19400011.log, p19400011.mc OK? ____

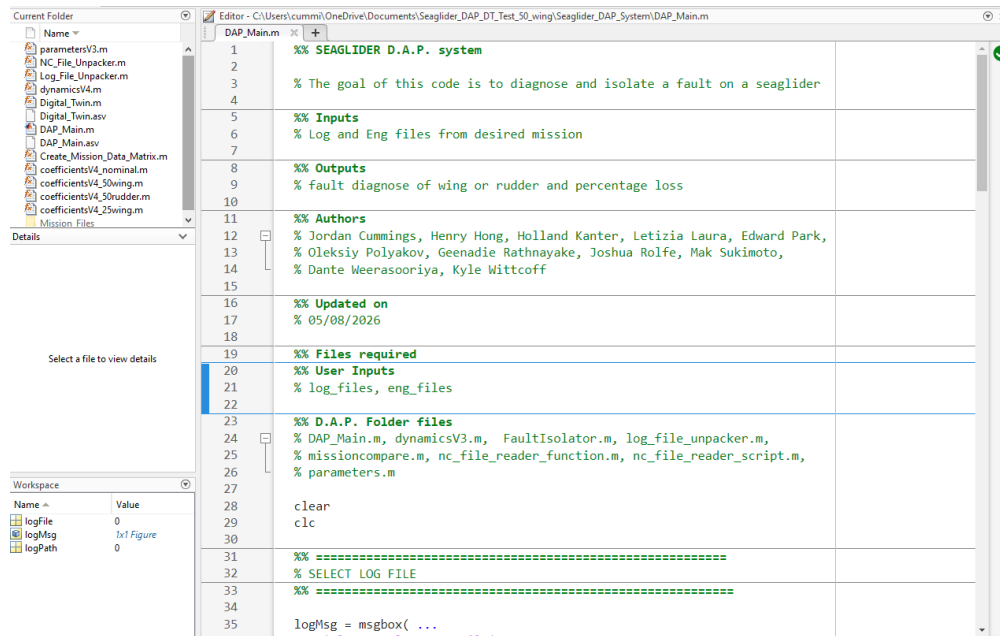


Figure 2: D.A.P. Main file

5.2. Running the Test code

The following outlines the procedure to run the D.A.P. system for the digital twin.

1. Click the run button (green right arrow) on the D.A.P main function to enable the function to run. OK? ____

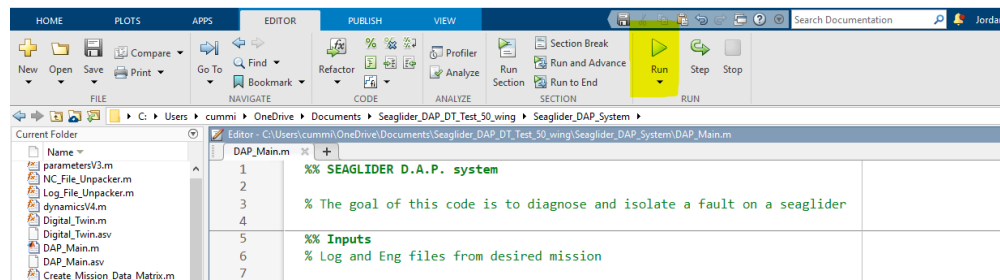


Figure 3: Running D.A.P. Main file

2. At the prompt for the Log file, Press 'OK' and your files window will open. Select the desired Log file from your computer. OK? ____

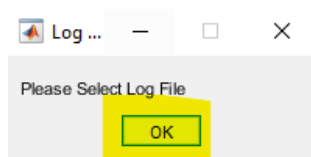


Figure 4: Pop-up window for LOG File selection

3. At the prompt for the NC file, Press 'OK' and your files window will open. Select the desired NC file from your computer. OK? ____

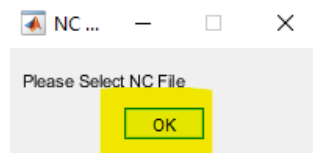


Figure 5: Pop-up window for NC File selection

4. The code should now run until it's completed.
5. After the code has stopped running, make the following checks:
- In the workspace tab, double click on the Mission_Data_Matrix_Compare Value cell. Check that all the values are filled in and there are no "NaN" values. OK? ____
 - In the workspace tab, double click on the Nominal_Sim_Matrix Value cell. Check that all the values are filled in and that there are no "NaN" values. OK? ____

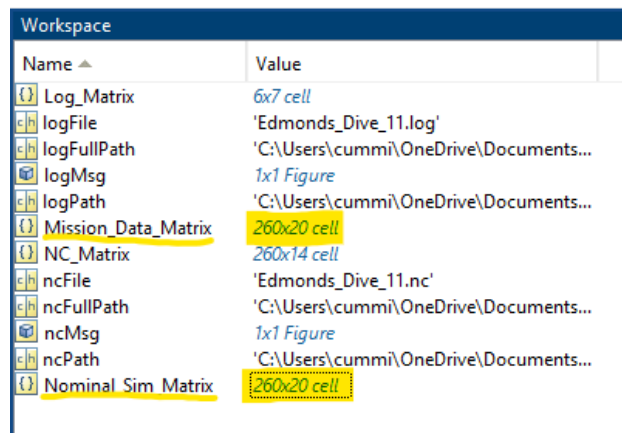


Figure 6: Workspace tab in MATLAB

6. Navigate to the graphs window and observe the graphs.
- Is the DT Depth within the error margin of the Mission Depth? YES/NO? ____
 - Is the DT Pitch within the error margin of the Mission Pitch? YES/NO? ____
 - Is the DT Roll within the error margin of the Mission Roll? YES/NO? ____
 - Is the DT Heading within the error margin of the Mission Heading?
YES/NO? ____
 - Is the DT Vertical Speed within the error margin of the Mission Vertical Speed?
YES/NO? ____

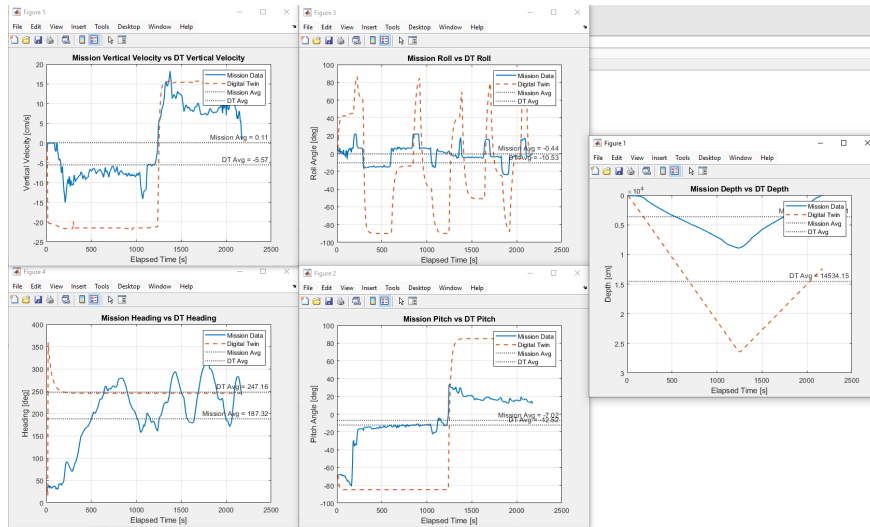


Figure 7: Graphs generated for Pitch, Roll, Heading, Vertical Velocity, and Depth

7. Under the file tab, save the graphs produced by the D.A.P function. OK? ____

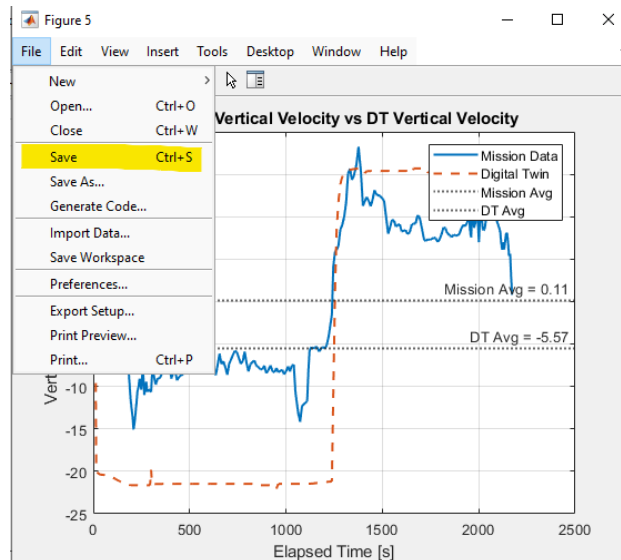


Figure 8: Save function for each graph

8. Rerun test code starting from subsection 5.2 with the next test data set.

6. Anomalies and Deviations

Document all unexpected behavior, procedure deviations, or hardware/software issues.

7. Shutdown Procedure

7.1. Shutdown

1. The D.A.P. main function will stop running once the task is finished.

8. Sign-Off

Test Conductor

Date

Reviewer / Witness

Date

Project Lead Approval

Date
